There are no four mutually unbiased bases in $\mathbb{C}^6$: a four-step proof

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Abstract

We prove that $\mathbb{C}^6$ admits no four mutually unbiased bases. The argument has four steps.

1. An exact algebraic fact about a four-variable seed polynomial R: its positive local maxima form a single orbit, with $\max R = 3^{13}/2^{12}$.
2. A topological consequence, relative to the classification of order-6 complex Hadamard matrices: the space of non-Tao Hadamard classes is connected.
3. A local exclusion lemma for balls around arbitrary Hadamard matrices.
4. A finite certified atlas of such balls whose boundaries transfer into one another.

Steps 1 and 2 are pure mathematics. Steps 3 and 4 are certified computation in interval arithmetic.

1 Setting

Hadamard matrices. Let $\mathcal{H}$ denote the order-6 complex Hadamard matrices: $|H_{ij}| = 1$ and $HH^* = 6I$. Write $H \cong H'$ if $H' = PD_1HD_2Q$, with P and Q permutation matrices and D_1 and D_2 diagonal unitaries.

Quartet edges. After a unitary change of frame, four mutually unbiased bases (MUBs) contain I and $H/\sqrt{6}$ for some $H \in \mathcal{H}$. Call H a *quartet edge* if $\{I, H/\sqrt{6}\}$ extends to four MUBs. This property is invariant under:

- equivalence;
- conjugation $H \mapsto \bar{H}$;
- transposition, since applying $H^*/\sqrt{6}$ exchanges the roles of I and $H/\sqrt{6}$.

We show that no $H \in \mathcal{H}$ is a quartet edge.

The quotient metric. A real 6×6 matrix is *horizontal* if all its row and column sums vanish. Let h be the orthogonal projector onto horizontal matrices. Its kernel consists of the row-plus-column patterns, so $\mathrm{e}^{iY} \cong \mathrm{e}^{i\mathrm{h}(Y)}$, with exponentials taken entrywise. On $\mathcal{H}/\!\cong$ we use

$$\delta(H, K) = \min\{\|v\| : \ v \text{ horizontal}, \ H \cong K \circ \mathrm{e}^{iv}\}.$$

Let S_6 be Tao's matrix and $X = (\mathcal{H}/\!\cong) \setminus \{[S_6]\}$.

Corner seeds. For $\theta \in \mathbb{T}^4$, let $E(\theta) = \left(\begin{smallmatrix} 1 & 1 & 1 \\ 1 & a & b \\ 1 & c & d \end{smallmatrix}\right)$ with $(a,b,c,d) = \mathrm{e}^{i\theta}$, and $G = 6I - EE^*$. Set $s = G_{21}$, $t = G_{31}$ and $r = G_{32}$, and form the invariants g, η, χ of the classification. Then

$$R(\theta) = 4g^2\eta - \chi^2.$$

On $S = \{R > 0\}$ an explicit formula produces a Hadamard completion $H_+(\theta)$ of $E(\theta)$. Relative to the classification, statements MC1–MC5 give

$$X = \{[H_+(\theta)]\} \cup \{[\overline{H_+(\theta)}]\} \cup \{[F_6]\} \qquad (\theta \in S).$$

2 Step 1: the maxima of R

Theorem 1. *The positive local maxima of R on $\mathbb{T}^4$ are exactly the* 18*-point orbit, under the* 144 *seed symmetries, of*

$$\theta_* = \alpha(1,2,-1,-2), \qquad \alpha = \arccos \tfrac{1}{4}, \qquad R(\theta_*) = 3^{13}/2^{12}.$$

The Hessian at θ_* is negative definite.

Outline. The deltoid traces. Normalise the even products (ad, b, c) and the odd products (bc, d, a) to have product 1, and let U and V be their traces. Each lies in the filled deltoid $\{\, Z : |Z|^4 + 18|Z|^2 - 8\,\mathrm{Re}\, Z^3 - 27 \le 0 \,\}$, and the trace map onto it is open.

The compact form of R. With $A = U + V$, $B = U - V$, $u = |A|^2$ and $v = |B|^2$, one has $g = 2v - 3u$, $\eta = v - 2u$ and

$$\chi = \tfrac{1}{2}(3u - v)(v - u + 12) - 2\,\mathrm{Re}(AB^2).$$

Thus R is invariant under the circle action $(A, B) \mapsto (\mathrm{e}^{-2i\tau}A, \mathrm{e}^{i\tau}B)$.

No interior maxima. In the double interior, criticality forces $AB^2 \in \mathbb{R}$. One resultant then leaves two real candidates, and at both $d^2R/d\psi^2 = -4\chi xv > 0$. A smooth boundary point paired with an interior point moves into the double interior along the circle action, by a curvature identity. A cusp is excluded directly.

The doubly boundary case. Here the stationary points are classified exactly; one spurious stationary value has positive transverse curvature.

All acceptance decisions are exact: resultants, Gröbner bases, Sturm counts and rational intervals. There are two independent proofs, in `proof/coverage/returns/`. □

3 Step 2: connectedness

Theorem 2. *Relative to MC1–MC5, the space X is connected.*

This upgrades the 2007 observation of Bengtsson et al. [2], that the then-known order-6 families form a connected set apart from S_6, to a theorem about all order-6 Hadamard matrices.

Proof. Sheet parity. If a seed symmetry L has transposition bit τ and conjugation bit κ, then

$$[H_\sigma(L\theta)] = \mathcal{C}^\kappa \mathcal{T}^\tau [H_{(-1)^{\tau+\kappa}\sigma}(\theta)].$$

The maximal frame $E(x) = \left(\begin{smallmatrix} 1 & 1 & 1 \\ 1 & x & x^2 \\ 1 & x^{-1} & x^{-2} \end{smallmatrix}\right)$ has two exact symmetries. Together these give $H_+(L\theta_*) \cong K_*$ for all 144 maps L.

Every component reaches $[K_*]$. The class map $\theta \mapsto [H_+(\theta)]$ is continuous on S, including across root cuts and sign switches. Every component of S has compact closure, with $R > 0$ inside and $R = 0$ on its boundary, so it contains a local maximum of R. By Theorem 1 that maximum lies in the orbit of θ_*. So $A = \{[H_+(\theta)]\}$ is connected and contains $[K_*]$, and likewise $\bar{A}$.

Joining at F_6. The family $\left(\begin{smallmatrix} F_3 & F_3 \\ F_3 D(t) & -F_3 D(t) \end{smallmatrix}\right)$, with $D(t) = \mathrm{diag}(1, \mathrm{e}^{it}, 1)$, tends to $F_2 \otimes F_3 \cong F_6$. Its cross-ratio e^{it} keeps it away from F_6 and S_6 for $0 < t < \pi/3$. So $[F_6]$ lies in the closure of both A and $\bar{A}$, and $X = A \cup \bar{A} \cup \{[F_6]\}$ is connected. □

Corollary 3. *Let* $\mathcal{R}$ *be a finite family of closed balls* $B_r = \{\mathrm{e}^{i(P_r+\delta)} : \delta \text{ horizontal}, \ \|\delta\| \le R_r\}$. *Let* W *be the set of classes of Hadamards in the union of its closure under* $\Gamma = \{\mathrm{id}$, *conjugation*, *transposition*, *conjugate-Suppose:*

1. W *contains a non-Tao Hadamard;*
2. *every Hadamard with* $\|\delta\| = R_r$, *for* $r \in \mathcal{R}$, *is equivalent after some* $g \in \Gamma$ *to one with* $\|\delta'\| < R_t$, *for some* $t \in \mathcal{R}$.

Then every class in X *meets* W.

Proof. $W \cap X$ is closed, as a finite union of compact images. It is open, since every point has an interior representative and δ is invariant under equivalence. It is nonempty. Since X is connected, $W \cap X = X$. □

4 Step 3: local exclusion

Let $\Phi(\delta) \in \mathbb{R}^{30}$ be the off-diagonal entries of $(\mathrm{e}^{i(P+\delta)})(\mathrm{e}^{i(P+\delta)})^*$, and $J = \Phi'(0)$. Entry (a, b) of $\Phi(\delta) - \Phi(0) - J\delta$ is $\sum_c Z_{ac}\bar{Z}_{bc}(\mathrm{e}^{iy_c} - 1 - iy_c)$ with $y_c = \delta_{ac} - \delta_{bc}$. Since $|\mathrm{e}^{iy} - 1 - iy| \le y^2/2$, $\|\delta_a - \delta_b\|^2 \le 2\|\delta\|^2$ and $\sum_{a<b} \|\delta_a - \delta_b\|^2 \le 6\|\delta\|^2$, we get, for every real δ and every centre,

$$\|\Phi(\delta) - \Phi(0) - J\delta\| \le \tfrac{L}{2}\|\delta\|^2, \qquad L = 2\sqrt{3}.$$

A *ball Record* is a rational centre P, a rational tangent frame T $(36 \times k)$, and a constant $\sigma > 0$ with $\|J\nu\| \ge \sigma\|\nu\|$ on the horizontal complement N of T. The constant σ is certified by an Arb Cholesky factorisation.

Lemma 4 (Normal bound)**.** *Every Hadamard* $\mathrm{e}^{i(P+\delta)}$ *with* $\|\delta\| \le R < \sigma/L$ *has* $\delta = T\eta + \nu$ *with* $\|\eta\| \le r_H$ *and* $\|\nu\| \le n$, *where*

$$n = \frac{\sigma - \sqrt{\sigma^2 - 2Lb}}{L}, \qquad b = \|\Phi(0)\| + \|JT\| r_H + \tfrac{L}{2}\lambda_{\max}(T^\top T)\, r_H^2.$$

Proof. We have $0 = \Phi(\delta) = \Phi(0) + J\delta + E$ with $\|E\| \le \frac{L}{2}\|\delta\|^2$, and $\|\delta\|^2 = \|T\eta\|^2 + \|\nu\|^2$. Hence $\sigma\|\nu\| - \frac{L}{2}\|\nu\|^2 \le b$. Since $\|\nu\| \le R < \sigma/L$, $\|\nu\|$ lies below the smaller root. □

Lemma 5 (Stage C, after [OAI, §6])**.** *Let* r_{col} *bound the columns of* δ *and* r_F *its Frobenius norm. Suppose the vector-bin cover, the label clustering, the O/U edge tests (including the joint test, which uses the common* η*) and the exhaustive two-clique search succeed with these bounds. Then no Hadamard in the ball is a quartet edge.*

All inequalities are proved in `proof/atlas/STAGEC.md`. Comparisons use Arb-tabulated exponentials and float arithmetic with margin 10^{-7}, against an error budget below 10^{-10}.

Remark 6 (S_6). At S_6 the horizontal spectrum of $J^\top J$ is exactly $\{9/2^{(16)}, 12^{(9)}\}$: an exact rational check gives $M(M - \frac{9}{2})(M - 12) = 0$ and the traces, and the kernel is the 11-dimensional gauge space. So $\sigma = 3/\sqrt{2}$ on all 25 horizontal directions ($k = 0$). Since $\Phi(0) = 0$, any Hadamard $\mathrm{e}^{i(P+\delta)}$ with δ horizontal satisfies $\sigma\|\delta\| \le \sqrt{3}\|\delta\|^2$, so $\delta = 0$ or $\|\delta\| \ge \sqrt{3/2}$. Thus S_6 is isolated. Stage C on the ball of radius 0.08 shows that S_6 is not a quartet edge.

5 Step 4: the atlas

Lemma 7 (Boundary transfer)**.** *Let s and t be Records, $\mathcal{A}$ a signed permutation of the 36 entries (row and column permutations, transposition, conjugation) and N an integer matrix. Put*

$$c = \mathrm{h}(\mathcal{A}P_s + 2\pi N - P_t), \qquad M = \mathrm{h}(\mathcal{A}T_s).$$

If $\sup_{\eta\in Q} \|c + M\eta\| + n_s < R_t$ on a box Q, then every boundary Hadamard of s with tangent coordinate in Q is equivalent to an interior point of t.

Proof. We have $\|\mathrm{h}(\mathcal{A}\nu)\| \le \|\nu\| \le n_s$, and representatives differ by row-plus-column patterns. Note also $\|M\| \le \|T_s\|$. □

The boundary therefore needs only a 4-dimensional cover. A box is either certified off the shell $R_s - n_s \le \|T\eta\| \le R_s + n_s$, or it transfers.

The computation. A breadth-first driver grows Records of radius 0.08 until every boundary transfers. An independent checker recomputes every Record from its centre phases: frame, σ, normal bound, Stage C, and the boundary tree with Arb transfer constants. The atlas has 12,020 Records, all certified, and F_6 lies in Record 877.

6 Conclusion

By Corollary 3 and Step 4, every class in X meets a Record. By Lemmas 4 and 5, no Hadamard in a Record is a quartet edge. By Remark 6, neither is S_6. Hence $\mathbb{C}^6$ has no four mutually unbiased bases. □

Inputs. The proof uses:

- the classification of order-6 complex Hadamard matrices [CT], through MC1–MC5, in Step 2 only;
- the Stage C method of [OAI], with every inequality reproved;
- certified computation in Steps 1, 3 and 4.

References

[1] M. Cárdenes Wuttig, J. Tindall, *A complete classification of complex Hadamard matrices of order six*, arXiv:2608.18053.

[2] I. Bengtsson, W. Bruzda, Å. Ericsson, J.-Å. Larsson, W. Tadej, K. Życzkowski, *Mutually unbiased bases and Hadamard matrices of order six*, J. Math. Phys. 48 (2007) 052106.

[3] OpenAI, *The maximum number of mutually unbiased bases in dimension six*, preprint (2026).